

Basics of Graphs

1.1. Graphs as Models

Social networks have become increasingly prominent in modern life. On the internet, a social network has many people as members. These people are friends with some people and not with others. This is a virtual version of real-life friendships. Instead of friendship, the relationship between people could be acquaintance or it could be biological relation.

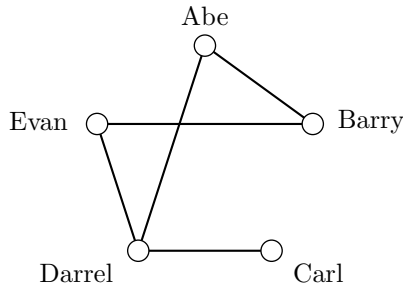
In other industries, more distinct relationships are possible. In the entertainment industry, we can ask whether two people have ever collaborated on some project. In academia, a natural relationship to consider is whether two people have ever coauthored a paper.

There are many questions we can ask about social networks. Who has the most friends? What is the largest number of people who are all friends? Given two people, what is the smallest number of people required to “connect” them?

Rather than try to solve equivalent problems separately in different contexts, we should use a mathematical model that can describe all of these situations. Then we can solve problems once in this abstract setting and apply the results to many different real-world problems.

We can model a social network by drawing a dot to represent each person and a line between two dots when two people are friends.

Example. Abe is friends with Barry and Darrel. Barry is friends with Abe and Evan. Carl is friends with Darrel. Darrel is friends with Abe, Carl, and Evan. Evan is friends with Barry and Darrel. These friendships are illustrated in the figure below.



The same model works just as well for other relationships between people, and many other real-world problems.

Definition 1.1. A **graph** G is a mathematical object consisting of a finite non-empty set of objects called **vertices** $V(G)$ (the **vertex set**), and a set of **edges** $E(G)$ (the **edge set**). An **edge** is two-element subset of the vertex set.

We commonly use G and H for graphs; $u, v, w, \dots$, for vertices; and e and f for edges. An edge $e = \{u, v\}$ will typically be written uv or vu , dropping the inconvenient braces.

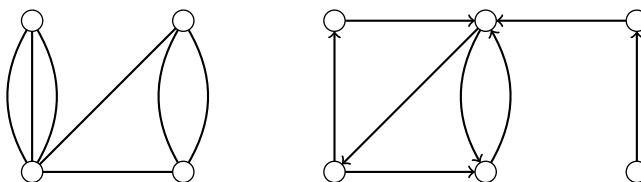
The name “graph” should not be confused with the graph of a function, an unrelated mathematical concept. The term “network”, which is common in computer science and technology, would probably be a more intuitive name. However, “graph” is now standard in mathematics. The terms “vertex” and “edge” come from geometry, as they can be used to represent the geometric objects with the same names.

Several variations on the concept of a graph are possible, allowing multiple edges or directed edges. If we allow multiple edges between vertices, the edges must be in a **multiset**, which allows multiple copies of the same object. For directed edges, we replace unordered pairs with ordered pairs of vertices.

Definition 1.2. A **multigraph** G is a mathematical object consisting of a finite nonempty set of objects called vertices $V(G)$ and a multiset $E(G)$ of pairs of vertices.

A **directed graph** (digraph) D is a mathematical object consisting of a finite nonempty set of objects called vertices $V(D)$ and a set $E(D)$ of ordered pairs of distinct vertices called **directed edges**.

A **directed multigraph** replaces the set $E(D)$ with a multiset.



Examples of a multigraph and a digraph are shown above. Every graph is also a multigraph. A multigraph is allowed, but not required, to have multiple edges between pairs of vertices.

Each of these mathematical objects can be used to model many real-world situations. Which one is chosen depends on whether multiple or directed edges make sense in the context of the problem.

Graphs can model transportation networks.

Example. A network of roads can be modeled using graph theory. Vertices represent intersections, and edges represent road segments. Often, a graph will be sufficient to model this situation. However, some areas have one-way streets, which should be represented by directed edges. Thus a digraph is the appropriate model in this situation. Sometimes there may be more than one road between the same pair of intersections. In this case, we would use a multigraph with multiple edges between some vertices.

Example. Airplane flights can be modeled with vertices representing airports. If edges represent flights, they are directed and likely multiple. Edges could alternatively represent the existence of a regular flight between two airports. In this case, the edges are not multiple.

Graphs also model communication and information networks.

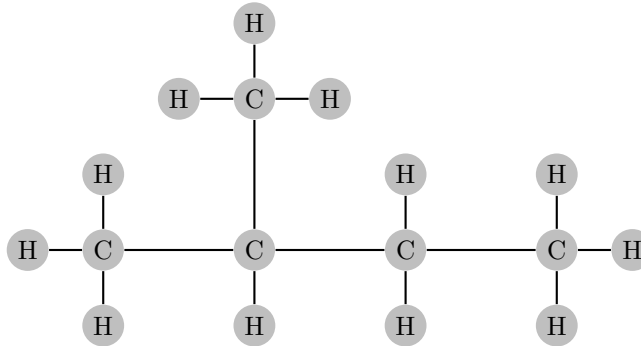
Example. A computer network has several computers connected by cables. A graph modeling this situation has vertices represent computers and edges represent cables.

Example. The **web graph** is a digraph that models the internet. Vertices represent webpages. A directed edge represents when one website links to another. The web graph is very large and growing. It changes as sites are added or deleted, and as links are added or deleted. Many of the other graphs modeling real-world situations also change over time.

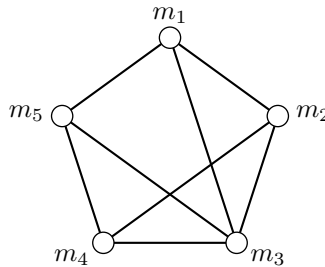
Example. A **citation graph** has vertices representing documents such as academic papers, patents, or legal opinions. A directed edge goes to the cited document from the document that cites it. Citations point toward documents further back in time. However, it is possible for two academic papers to both cite each other if they were written by the same author or if their authors corresponded while they were being written.

Graphs can model topics as diverse as chemical molecules and scheduling meetings.

Example. A chemical molecule has several atoms that are held together by chemical bonds. A graph models this situation with vertices for atoms and edges for chemical bonds. Vertex and edge labels may be necessary to identify different atoms and different types of bonds. A graph representing isopentane, C_5H_{12} , is shown below.



Example. Eight businessmen must have five meetings. The sets of people in meetings are $m_1 = \{1, 2, 3, 4\}$, $m_2 = \{1, 5, 6\}$, $m_3 = \{2, 5, 7\}$, $m_4 = \{6, 7, 8\}$, and $m_5 = \{3, 4, 7, 8\}$. We draw a graph where vertices represent meetings and there is an edge between vertices when the sets have a nonempty intersection. We would like to schedule the meetings in as few time slots as possible. Observation shows that meetings 1 and 4 can be scheduled in one slot, 2 and 5 in another, and 3 in a third. Meetings 1, 2, and 3 all need separate slots, so three slots are required. The graph in this example is called an **intersection graph**, which has vertices represent sets and edges between vertices when the sets have a nonempty intersection.



The situations that can be modeled with graph theory are seemingly endless. You can even construct a graph whose vertices represent other graphs! Many more situations that can be modeled with graph theory are explored in the Exercises.

1.2. Representations of Graphs

There are several terms associated with a graph containing an edge $e = uv$.

Definition 1.3. If $e = uv$ and $f = uv$ are edges of a graph, then we write $u \leftrightarrow v$, and we say u and v are **adjacent**, u and v are **neighbors**, u is **adjacent to** v , e **joins** u and v , u is **joined to** v , e and v are **incident**, and e and f are **adjacent**. If u and v are not adjacent, we write $u \nleftrightarrow v$, and we say u and v are **nonadjacent**.

We can describe a graph by simply listing the vertex and edge sets. While this description is accurate, it tends not to be particularly helpful for solving problems or making new discoveries. There are alternative ways of describing graphs.

Definition 1.4. An **adjacency list** of a graph is a list of each vertex in a separate row followed by a list of the vertices that it is adjacent to. An **adjacency matrix**

of a graph G with vertex set $\{v_1, \dots, v_n\}$ is an $n \times n$ matrix with i, j entry 1 when $v_i v_j \in E(G)$, and 0 otherwise.

The adjacency matrix of a graph is symmetric and has 0's on the diagonal.

Example. Let G be the graph with vertex set $V(G) = \{1, 2, 3, 4, 5\}$ and edge set $E(G) = \{12, 13, 23, 24, 34, 45\}$. The adjacency list and adjacency matrix are given below.

Vertex	Adjacent to
1	2, 3
2	1, 3, 4
3	1, 2, 4
4	2, 3, 5
5	4

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

For sparse graphs (those with relatively few edges), an adjacency list uses less space than an adjacency matrix. For dense graphs (those with many edges), the space usage is comparable. Each has advantages and disadvantages for computational efficiency. However, the adjacency matrix has more theoretical uses, since the theory of linear algebra can be applied. For example, in Section 1.6, it is used to count walks in graphs.

It is often beneficial to represent a graph using a drawing. Indeed, much of the appeal of graph theory comes from fact that graphs can be drawn and analyzed visually.

Definition 1.5. A **drawing of a graph** is a diagram with a small circle (either open or solid) in the plane representing each vertex and a curve (often a straight line) joining two circles representing each edge.

The names of the vertices can be written near them in the drawing. Less commonly, the names of edges can also.

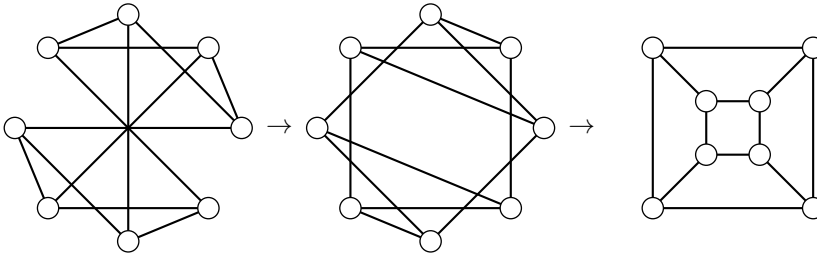
But how should a graph be drawn? At a minimum, vertices must have distinct locations, and edges must not intersect vertices to which they are not incident. Edges may cross, but usually they should not cross more than once. But there are still (infinitely) many ways to draw a graph. We may prefer a drawing that has other desirable properties, such as no edge crossings (if possible), reflective or rotational symmetry, and no long edges. There is no one best drawing; different drawings may emphasize different properties of a graph. The field of **graph drawing** studies how best to visually represent graphs.

One way to satisfy the essential conditions above is to place all the vertices on a circle and make all the edges straight lines that are chords of the circle. Some computer programs use this method when not given any information about where to place the vertices. This method tends to lead to many edge crossings and long edges, however.

Another possibility is to start with one vertex, draw its neighbors close to it, try to draw their neighbors close to them, and so on. No matter what method is used, it usually takes several attempts to find a desirable drawing. An undesirable

drawing can be improved by moving one or more vertices to another location and repeating this step until a better drawing is found.

Example. Suppose we draw the graph below left, with its vertices initially around a circle. We notice that the graph contains several 4-cycles. We swap two pairs of vertices to unravel two 4-cycles, obtaining the middle drawing. Finally, we move one 4-cycle inside the other so that no edges cross, obtaining the drawing at right. We obtain a symmetric, visually pleasing drawing.



The fact that a graph can be drawn in many different ways raises the issue that two graphs that appear to be different may be essentially the same. One way this may be is that the vertex sets are different, but the vertices can be renamed so that the graphs are the same. Equivalently, the graphs can be drawn identically except for the vertex names. Another possibility is that the vertices may have the same names, but the graphs may be drawn differently.

We say that two such graphs are **isomorphic**. Loosely speaking, this means that the vertices can be renamed or moved so that the graphs are the same. It is usually not too hard to tell whether or not two graphs are the same, but finding a method that always works is a difficult problem. We consider this problem in depth in Section 3.2, where we also state a precise definition of **isomorphism**.

It is possible to draw the diagram of a graph without naming its vertices. We call this an **unlabeled graph**. An unlabeled graph can be viewed as representing all possible labeled graphs with the same structure. Two unlabeled graphs are isomorphic if it is possible to label their vertices so that they are the same. If two unlabeled graphs G and H are isomorphic, we write $G = H$.

We will refer to a graph with named vertices as a **labeled graph**. Without additional context, “graph” will mean an unlabeled graph. Practical applications usually involve labeled graphs, while theoretical problems more commonly involve unlabeled graphs. When we work with a labeled graph that does not derive from a specific application, we simply need n distinct names for the vertices, where n is the number of vertices in the graph. The natural choice is to use the set $[n] = \{1, \dots, n\}$ as the vertex set.

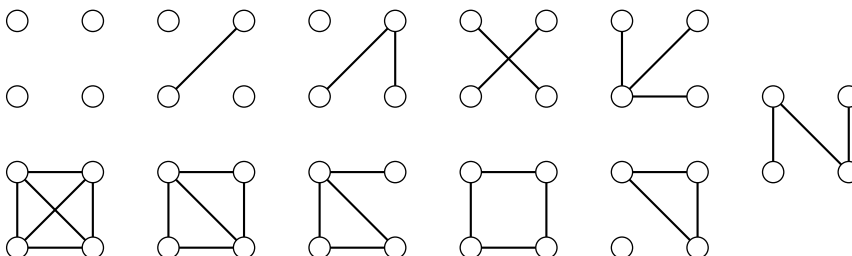
Example. Suppose we want to count the number of labeled graphs with vertex set $[n]$. There are $\binom{n}{2} = \frac{n(n-1)}{2}$ pairs of vertices, each of which may be joined by an edge or not. There are two possibilities for each pair, and the choices are independent. Thus there are $2^{\binom{n}{2}}$ labeled graphs with vertex set $[n]$.

Basic counting techniques, including those just used, are discussed in Section 2 of the Appendix.

There are $n!$ ways to permute the vertices of a labeled graph. Thus there are at least $\frac{2^{\binom{n}{2}}}{n!}$ unlabeled graphs. This is not exact since the number of labeled graphs corresponding to each unlabeled graph varies. Using generating functions, it is possible to generate the sequence of the number of unlabeled graphs with n vertices: 1, 2, 4, 11, 34, 156, 1044, 12346, ... (sequence A000088 in the Online Encyclopedia of Integer Sequences (OEIS)). This sequence is asymptotic to $\frac{2^{\binom{n}{2}}}{n!}$.

Example. To find all unlabeled graphs with four vertices, we could just start drawing graphs. But then we could not be certain that we had found all of them. This is one of many problems in graph theory that require a careful “case-checking” argument to be certain that all solutions have been found. We need to break down the possibilities using the properties of the graphs.

With four vertices, the number of edges must be between 0 and 6. There is only one graph when the number of edges is 0, 1, 5, or 6. When there are two edges, they can either be adjacent or not. When there are four edges, the two missing edges can either be adjacent or not. When there are three edges, they could all be incident with the same vertex. When two are adjacent, the third can either be adjacent to both of their other ends, or only one. Thus there are 11 graphs with four vertices (graphed below), and we can be confident we have them all.



1.3. Graph Parameters

We need simple notation for the number of vertices and edges of a graph.

Definition 1.6. The **order** $n(G) = |V(G)|$ of a graph G is the number of vertices of G . The **size** $m(G) = |E(G)|$ of a graph G is the number of edges of G .

When the context is clear (typically when there is a single graph under discussion), we will simplify the notation for order and size to n and m , respectively. We will only use n and m to represent order and size (thus the complete graph K_n has order n), but caution is still warranted. For example, the statement “ $n(K_n - v) = n - 1$ ” should be revised to “ $n(K_r - v) = r - 1$ ” or “the order of $K_n - v$ is $n - 1$.”

We note in passing that graph theory is a relatively young subject, and its notation is still evolving. Most terminology and notation have changed from past texts and papers. Some notation is still not standardized (for example, the notations p , $|G|$, n , $n(G)$, and $|V(G)|$ are all common for order). Other terms are now standard but have varied in the past. To avoid unnecessary confusion, this text

will usually not mention alternate terminology or notation, but readers are urged to use caution when reading other sources.

Order and size are examples of functions that are defined on all graphs.

Definition 1.7. A **graph parameter** $f(G)$ is a function that is from some or all graphs to the real numbers.

Most parameters of interest have integer values. Thinking of parameters as functions is not especially beneficial in graph theory. For one thing, parameters of interest are almost never one-to-one, so inverse functions usually do not exist.

Some parameters are defined on the vertices of a particular graph.

Definition 1.8. The **degree** of a vertex v , written $d_G(v)$ or $d(v)$ when the graph in question is clear, is the number of edges incident with v . A vertex with degree 0 is an **isolated vertex**. A vertex of degree 1 is a **leaf**. An **even vertex** has even degree; an **odd vertex** has odd degree. The **neighborhood** of a vertex v , $N(v)$, is the set of neighbors of v .

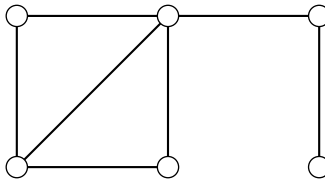
Note that the degree of a vertex is the number of vertices in its row of the adjacency list, and the number of 1's in its row (or column) of the adjacency matrix.

Example. Airlines usually do not fly directly between two smaller cities. Instead, they fly from a smaller city airport to a hub airport, which has flights to many other smaller airports and other hub airports. A hub airport can be identified by having a relatively large degree in a graph representing flights.

We can also consider the degrees of an entire graph.

Definition 1.9. The **degree sequence** of a graph G is the list of its degrees, usually written in nonincreasing order. Its **minimum degree** is $\delta(G)$. Its **maximum degree** is $\Delta(G)$. It is **regular** if every vertex has the same degree (k -**regular** if the common degree is k). A 3-regular graph is a **cubic graph**.

Example. The following graph G has degree sequence 4, 3, 2, 2, 2, 1, $\Delta(G) = 4$, and $\delta(G) = 1$.



There are many relationships among graph parameters. The following theorem is basic enough that it is called the First Theorem of Graph Theory.

Theorem 1.10 (First Theorem of Graph Theory). If G is a graph, then $\sum d(v_i) = 2m$.

Proof. Consider the set of all vertex-edge incidences (the “ends” of edges) in a graph. Partitioning the set by vertices shows its cardinality is the sum of the degrees. Partitioning the set by edges shows each edge appears twice, so its cardinality is $2m$. Thus $\sum d(v_i) = 2m$. $\square$

This theorem is also known as the **degree sum formula**. The proof uses the technique known as *counting two ways*, which partitions a set two different ways and counts both ways to obtain an identity. This technique is explored in detail in Section 2 of the Appendix on counting techniques.

Corollary 1.11. *Every graph has an even number of vertices of odd degree.*

Proof. Assume to the contrary that a graph has an odd number of odd vertices. Then its degree sum must be odd, which contradicts the First Theorem. $\square$

Both this corollary and the First Theorem are sometimes called the **Handshaking Lemma**. The results are interpreted involving people at a party shaking hands. With vertices representing people and edges representing handshakes, it follows that an even number of people shake an odd number of hands.

Several other observations follow immediately from the First Theorem. The average degree of a graph is $\frac{2m}{n}$. Hence $\delta(G) \leq \frac{2m}{n} \leq \Delta(G)$. Also, the size of a k -regular graph with order n is $m = \frac{nk}{2}$.

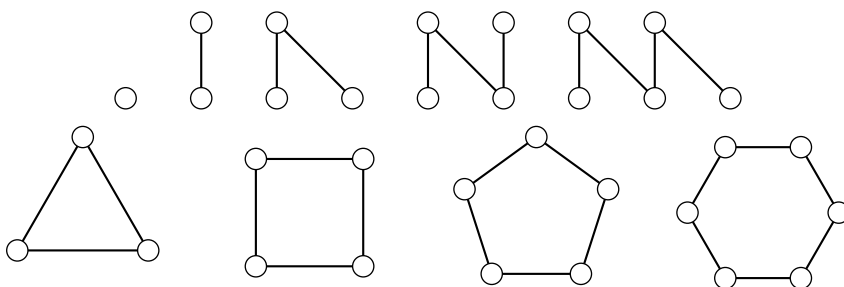
1.4. Common Graph Classes

Definition 1.12. A **graph class** is a set of graphs. We denote a graph class using blackboard bold ($\mathbb{G}$ or $\mathbb{H}$) if it does not have its own notation.

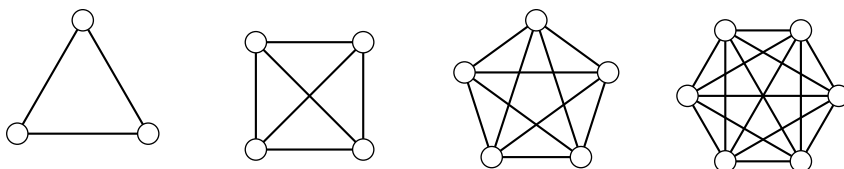
We now introduce several important graph classes.

Definition 1.13. A **path** P_n is a graph whose vertices can be numbered $v_1, v_2, \dots, v_n$ so that its edges are $v_1v_2, \dots, v_{n-1}v_n$. A **cycle** C_n (or n -cycle) is a graph whose vertices can be numbered $v_1, v_2, \dots, v_n$ so that its edges are $v_1v_2, \dots, v_{n-1}v_n$, and v_nv_1 . An **even cycle** has n even, and an **odd cycle** has n odd.

Small paths and cycles are illustrated in the following figures.



Definition 1.14. A **complete graph** K_n has order n and every pair of vertices is adjacent. An **empty graph** $\bar{K}_n$ has order n and no edges. The (unique) graph with one vertex K_1 is called the **trivial graph**. A graph with more than one vertex is **nontrivial**.



The complete graph K_n has size $\binom{n}{2} = \frac{n(n-1)}{2}$ since every pair of vertices produces an edge. Complete and empty graphs are opposite extremes for the size of a graph with n vertices.

These graph classes are interesting in their own right. When we study an unfamiliar parameter or property of graphs, we usually apply it to these classes first, since they are familiar and easy to define. However, these classes are also important because they are contained in other graphs.

Many mathematical objects contain subobjects. Sets contain subsets. Groups contain subgroups. Vector spaces contain subspaces. The same is true for graphs.

Definition 1.15. A graph H is a **subgraph** of a graph G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$ so that the edges in $E(H)$ use only vertices in $V(H)$. We write $H \subseteq G$ and say G **contains** H . An **induced subgraph** is a subgraph $G[S]$ with vertex $S \subseteq V(G)$ and all edges with both ends in S .

Equivalently, an induced subgraph can be obtained by deleting a set of vertices and all edges incident with them.

Definition 1.16. A **clique** is a complete subgraph, or the set of vertices inducing a complete subgraph. An **independent set** of vertices is a set that induces an empty graph.

Both of these concepts are used heavily in graph coloring and elsewhere in graph theory.

Graphs that do not contain certain subgraphs are also of interest.

Definition 1.17. A graph G is **H -free** if it does not contain any induced subgraph isomorphic to H .

The graph K_3 is called a **triangle**. A K_3 -free graph is called **triangle-free**.

Paths are relevant as subgraphs when discussing connectivity and distance (Section 1.6). Cycles are relevant as subgraphs in Eulerian and Hamiltonian graphs, planar graphs, and more.

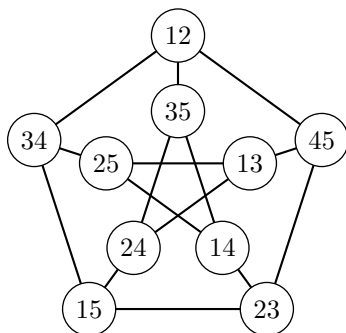
Definition 1.18. A graph, subgraph, or structure of a graph is **maximal** if no larger one contains it. It is **maximum** if it is as large as possible. The definitions of **minimal** and **minimum** are similar.

Lemma 1.19. *If a graph G has $\delta(G) \geq 2$, then it contains a cycle.*

Proof. Let P be a maximal path in G with vertices $v_1, \dots, v_n$. Now v_n has degree at least 2, so it is adjacent to some vertex $u \neq v_{n-1}$. If u is not on the path, it could be extended, contradicting its maximality. Thus u must be on the path, forming a cycle. $\square$

The distinction between *maximal* and *maximum* is important. Any maximum structure is maximal, but the converse may not be true. A maximal path may not be maximum. The same is true for cliques and independent sets.

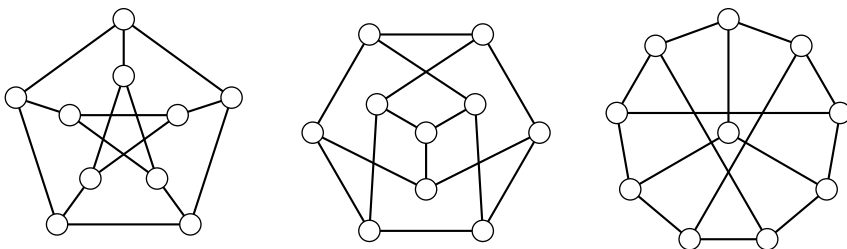
Definition 1.20. The **Petersen graph** has vertices that represent the 2-element subsets of $[5] = \{1, 2, 3, 4, 5\}$. Edges join vertices whose subsets are disjoint.



The Petersen graph is a particularly important example in graph theory. It has several surprising properties that will be revealed throughout this text. In fact, there is an entire book about it (Holton/Sheehan [1993])!

There are $\binom{5}{2} = 10$ 2-element subsets of $[5]$, so the Petersen graph has order 10. There are three elements of $[5]$ not contained in a given 2-element subset, so each vertex is adjacent to three others. Thus the Petersen graph is cubic, so it has size $\frac{10 \cdot 3}{2} = 15$. There cannot be three disjoint 2-element subsets of $[5]$, so the Petersen graph is triangle-free.

It can be drawn to emphasize two 5-cycles with vertices 12, 34, 51, 23, 45 and 13, 52, 41, 35, 24, respectively, with edges matching vertices on each cycle. Other drawings of the Petersen graph emphasize other interesting properties.



Proposition 1.21. *Two nonadjacent vertices of the Petersen graph have exactly one common neighbor. Thus the Petersen graph has no 4-cycle.*

Proof. Nonadjacent vertices have sets with one element in common. Thus there are two elements of $[5]$ in neither of them, which produces one vertex adjacent to both. A 4-cycle would require nonadjacent vertices with two common neighbors. $\square$

Definition 1.22. The **girth** of a graph is the length of its shortest cycle.

Thus the Petersen graph has girth 5.

Counting different types of graphs or subgraphs is a common problem in graph theory. The following result uses the counting technique known as counting by bijection (Appendix, Section 2).

Proposition 1.23. *The Petersen graph contains 15 8-cycles.*

Proof. Any 8-cycle omits two vertices. They must be adjacent, or else they would have a common neighbor that could not be on the cycle. Deleting two adjacent

vertices from the Petersen graph results in a graph that clearly contains a single 8-cycle. The Petersen graph has 15 edges and, hence, 15 pairs of adjacent vertices, so it has 15 8-cycles. $\square$

There are several classes of graphs that generalize the Petersen graph, which are explored in the Exercises.

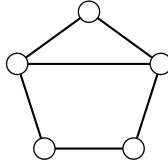
1.5. Graph Operations

Almost any type of mathematical object has corresponding mathematical operations to manipulate the objects and produce new objects. Numbers can be negated, added, or divided. Functions can be multiplied or composed. Sets can be complemented or intersected. Thus it is natural that analogous operations exist for graphs.

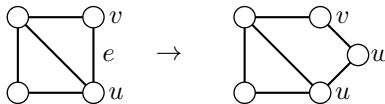
A **unary operation** produces a new graph when given a single graph.

Definition 1.24. The graph $G - e$ or $G - X$ is obtained by deleting edge e or edge set X from $E(G)$. The graph $G - v$ or $G - S$ is obtained by deleting vertex v or vertex set S and all incident edges from G . The graph $G + e$ is obtained by adding e to the edges of G .

When G is an unlabeled graph, $G - e$ will only be defined when a particular edge is specified or when deleting any edge produces the same graph. For example, $C_n - e = P_n$ no matter which edge is deleted. The same will hold for vertices. When e is unspecified, $G + e$ will only be defined when any choice yields the same graph. The graph C_5 satisfies this condition, so $C_5 + e$ is defined.



Definition 1.25. A **subdivision of an edge** $e = uv$, deletes e and adds vertex w and edges uw and wv . A graph H is a **subdivision of a graph** G if it can be obtained by some number (perhaps zero) of subdivisions of edges of G .



Loosely speaking, a subdivision can be thought of as inserting a vertex on an edge. Subdivisions are important when we are mainly interested in whether there is a path between vertices of a graph, regardless of length. This includes connectivity (Section 2.3) and planarity (Section 5.2). Note that a subdivision of an edge increases the order and size by one.

The preceding operations can be considered “local” operations that leave most of the graph unchanged. Unary operations can also change the entire graph.

Definition 1.26. The **complement** $\overline{G}$ of a graph G has the same vertex set and edge $uv \in E(\overline{G})$ if and only if $uv \notin E(G)$. A **decomposition** of G is a set of

nonempty subgraphs whose edge sets partition $E(G)$. The subgraphs are said to **decompose** G .

Note that an empty graph is the complement of a complete graph of the same order (and vice versa), explaining the notation $\overline{K}_n$ for empty graphs. A graph G of order n and its complement decompose K_n .

Example. The complement of C_4 is $2K_2$, so $\{C_4, 2K_2\}$ is a decomposition of K_4 .

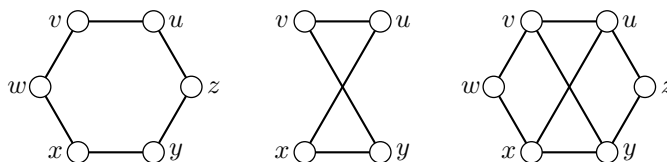
A unary operation may produce a graph with a completely different vertex set. An example of this is the line graph (Section 2.4).

A **binary graph operation** uses two graphs to produce a new graph. We consider several binary operations: the union, join, and Cartesian product of graphs.

Definition 1.27. The **union of graphs** G and H , $G \cup H$, is the graph with vertex set $V(G) \cup V(H)$ and edge set $E(G) \cup E(H)$. The union is a **disjoint union** if the vertex sets of the graphs are disjoint. A disjoint union of k copies of G is denoted kG .

When we work with unlabeled graphs, we consider a union disjoint.

Example. The union of the two graphs at left is the graph at right.



Empty graphs, complete graphs, and cycles are all regular. In fact, 0-regular graphs are exactly the empty graphs, which can also be denoted nK_1 . Any 1-regular graph must be composed of disjoint copies of K_2 , that is kK_2 . Next we characterize the structure of 2-regular graphs.

Proposition 1.28. Any 2-regular graph is a disjoint union of cycles.

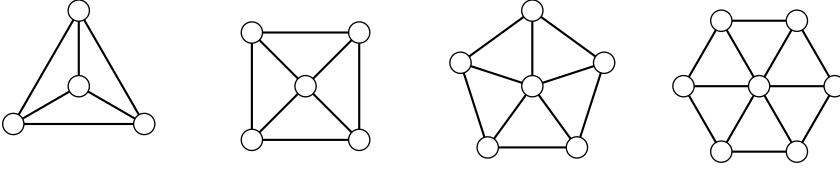
Proof. We use strong induction on the number of cycles of a 2-regular graph G . By Lemma 1.19, G must contain a cycle. If G is a single cycle, we are done. Assume the result holds for graphs with fewer than k cycles. Let G contain k cycles, one of which is C . All of the vertices of C have degree 2 in G . Deleting these vertices produces a 2-regular graph H with fewer than k cycles. Thus H must be a disjoint union of cycles, and hence so is G . $\square$

Strong induction is a common proof technique in graph theory. Section 1 of the Appendix reviews common proof techniques.

The structure of k -regular graphs is much more complicated when $k \geq 3$. This problem is considered further in later sections.

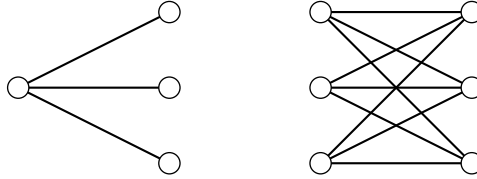
Definition 1.29. The **join of graphs** G and H , $G + H$, is obtained from the disjoint union $G \cup H$ by adding the edges $\{uv : u \in V(G), v \in V(H)\}$.

The **wheel** W_n , $n \geq 4$, is the join of a cycle and a single vertex, $W_n = C_{n-1} + K_1$.



Definition 1.30. The **complete bipartite graph** $K_{r,s}$ is the join of two empty graphs, $K_{r,s} = \overline{K}_r + \overline{K}_s$. A **star** is the complete bipartite graph $K_{1,s}$. The star $K_{1,3}$ is known a **claw** when it is an induced subgraph of another graph.

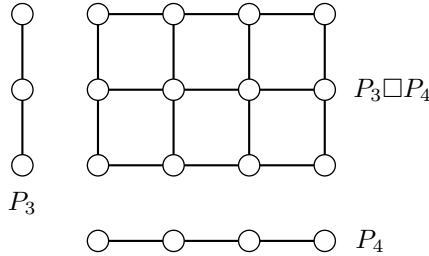
General (noncomplete) bipartite graphs will be defined in Section 1.7. The graphs $K_{1,3}$ and $K_{3,3}$ are shown below.



Definition 1.31. The **Cartesian product** of G and H , $G \square H$, has vertex set $V(G) \times V(H)$ and (u, v) adjacent to (u', v') if $u = u'$ and $vv' \in E(H)$ or $v = v'$ and $uu' \in E(G)$.

The notation $G \square H$ symbolizes the fact that $K_2 \square K_2 = C_4$. Note that $G \square H$ has a copy of G for each vertex of H , and vice versa. Thus it decomposes into copies of G and H .

Example. The Cartesian product of P_3 and P_4 is shown below.



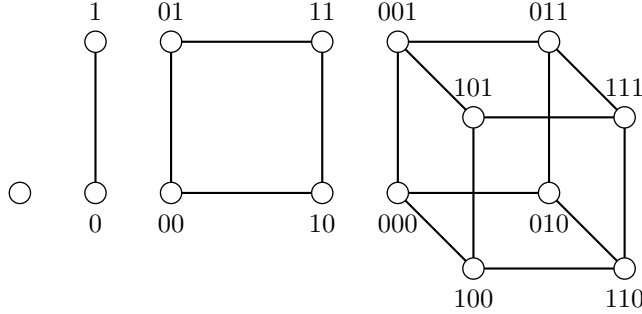
Definition 1.32. The **grid** $G_{r,s}$ is the Cartesian product of two paths, $G_{r,s} = P_r \square P_s$.

The definitions of the union, join, and Cartesian product all extend in the natural way to more than two graphs. (Note that they are all associative and commutative operations.) This can be used to define another important class of graphs.

Definition 1.33. The **hypercube** Q_k is defined recursively by $Q_1 = K_2$ and $Q_k = Q_{k-1} \square K_2$.

Note that $Q_2 = C_4$ and Q_3 represents the usual three-dimensional cube. The hypercube Q_k can also be defined as a graph with vertices representing k -digit

bitstrings that are adjacent when they differ in a single digit. Note that Q_k is k -regular and has order $n(Q_k) = 2^k$. Thus $m(Q_k) = \frac{k \cdot 2^k}{2} = k \cdot 2^{k-1}$.



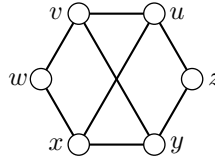
1.6. Distance

We now consider how we can travel around a graph via adjacent edges.

Definition 1.34. A **walk** is a list $v_0, e_1, v_1, \dots, e_k, v_k$ of vertices and edges such that $e_i = v_{i-1}v_i$. A $u-v$ walk has $u = v_0$ and $v = v_k$. These are its **endpoints**. A walk is **closed** if its endpoints are the same. A $u-v$ **path** is a path with endpoints u and v . Its other vertices are **internal vertices**. The **length** of a walk, path, or cycle is the number of edges it contains.

Note that a path is a walk with no repeated vertices. The length of P_n is $n-1$.

Example. In the graph below, u, z, y, v, u, x, y, v is a $u-v$ walk of length 7, and u, x, y, v is a $u-v$ path of length 3.



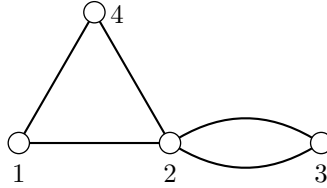
The adjacency matrix can be used to count walks and some cycles.

Theorem 1.35. Let G be a multigraph with vertices $v_1, \dots, v_n$ and adjacency matrix $A(G)$. Then the i, j entry of A^k is the number of $v_i - v_j$ walks of length k in G .

Proof. We use induction on k . For $k = 1$, the result is immediate since a walk of length 1 contains a single edge. Assume for some k that the i, j entry of A^k , $a_{ij}^{(k)}$, is the number of $v_i - v_j$ walks of length k in G . Now $A^{k+1} = A^k A$, so $a_{ij}^{(k+1)} = \sum_{t=1}^n a_{it}^{(k)} a_{tj}$. Now every $v_i - v_j$ walk of length $k+1$ contains a $v_i - v_t$ walk of length k and an edge from v_t to v_j . Thus the summation counts the number of $v_i - v_j$ walks of length $k+1$. $\square$

Example. In the movie Good Will Hunting, a four-part problem is presented. For the graph below, the first two parts ask for

1. the adjacency matrix A ,
2. the matrix giving the number of 3-step walks.



The adjacency matrix A is below left. The matrix of 3-step walks is just A^3 .

$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 2 & 1 \\ 0 & 2 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}, \quad A^3 = \begin{bmatrix} 2 & 7 & 2 & 3 \\ 7 & 2 & 12 & 7 \\ 2 & 12 & 0 & 2 \\ 3 & 7 & 2 & 2 \end{bmatrix}$$

The movie portrays the problem as being so difficult that only a genius like Will Hunting can solve it. But the first two parts are easy. The last two parts, which involve generating functions, are harder, but not that hard.

Definition 1.36. The **trace** of a matrix A , $\text{tr}(A)$, is the sum of the entries on its forward diagonal.

Corollary 1.37. The number of triangles in a graph G with adjacency matrix A is $\frac{1}{6} \text{tr}(A^3)$.

Proof. A triangle is a closed walk of length 3. These walks are counted on the forward diagonal of A^3 . Each triangle can be counted six ways, since there are three starting vertices and two directions. Thus $\frac{1}{6} \text{tr}(A^3)$ is the number of triangles in G . $\square$

Paths are important subgraphs because they are used to determine whether a graph is all one piece or not.

Definition 1.38. A graph is **connected** if it contains a $u - v$ path for all vertices u and v . A graph is **disconnected** if it is not connected. A **component** of a graph is a maximal connected subgraph.

Any disconnected graph is a disjoint union of its components. For example, the components of $C_4 \cup P_3$ are C_4 and P_3 .

Many graph parameters can be determined by calculating them over each component and then taking the maximum, minimum, or sum. For example, the size of a graph is the sum of the sizes of its components, and the maximum degree of a graph is the maximum of the maximum degrees of its components.

When a graph is connected, we would like to know how far apart its vertices are.

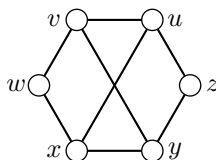
Definition 1.39. The **distance** between vertices u and v , $d(u, v)$, is the length of the shortest $u - v$ path. If u and v are in distinct components, we say $d(u, v) = \infty$. A $u - v$ **geodesic** is a $u - v$ path of length $d(u, v)$. The **diameter** of a graph G , $\text{diam}(G)$, is the maximum length of a geodesic in G . The **eccentricity** of a vertex

$e(v)$ is the maximum length of a geodesic starting at v . The **radius** of a graph G , $\text{rad}(G)$, is the minimum eccentricity of its vertices. The **center** of a graph is the subgraph induced by the vertices of minimum eccentricity.

Example. The social network that represents collaboration in Hollywood is called the **Hollywood graph**. The **Bacon number** of an actor is the distance between that actor and Kevin Bacon in the Hollywood graph. Bacon's own Bacon number is 0. Tom Hanks's Bacon number is 1, since they appeared together in Apollo 13. Matt Damon's Bacon number is 2, since Damon has not worked with Bacon, but appeared with Hanks in Saving Private Ryan. In the game "**Six Degrees of Kevin Bacon**", the goal is to find a geodesic path from some actor to Bacon. It has become a common party game and internet meme.

Example. The **academic collaboration graph** has an edge between two academics who are coauthors of a paper. The **Erdos number** of a mathematician (or other academic) is the distance between him and Paul Erdos in the academic collaboration graph. Erdos (1913–1996) was an iconic mathematician who solved many problems in graph theory, number theory, and other areas. Erdos is famous due to his eccentric lifestyle and the record number of papers he published, most with coauthors. This author's Erdos number is 2, as I coauthored a paper with Allen Schwenk, who coauthored with Erdos.

Example. In the following graph, $d(u, v) = 1$, the radius is 2, the diameter is 3, and the center is C_4 , induced by u, v, y , and x .



Example. The hypercube Q_k has diameter k , since changing all k coordinates requires k edges. The wheel $W_{n+1} = C_n + K_1$ has radius 1 and diameter 2. The center is the vertex corresponding to K_1 .

The definition of a path resembles the definition of a walk. However, a path cannot repeat vertices or edges.

Lemma 1.40. *Every $u - v$ walk contains a $u - v$ path.*

Proof. Assume to the contrary that there is a $u - v$ walk that does not contain a $u - v$ path, and let W be such a walk with minimum length. Then W must repeat some vertex w . Deleting all the vertices and edges on the walk between the occurrences of w produces a shorter $u - v$ walk. Either it is a $u - v$ path or it is a smaller counterexample. Either way, there is a contradiction. $\square$

Distance satisfies the triangle inequality $d(u, w) \leq d(u, v) + d(v, w)$ since concatenating the $u - v$ and $v - w$ paths produces a $u - w$ walk, which contains a $u - w$ path. Along with the obvious properties $d(u, v) \geq 0$ with equality exactly when $u = v$, and $d(u, v) = d(v, u)$, this shows that graphs with distance are a metric space.

Proposition 1.41. *For every connected graph, $\text{rad}(G) \leq \text{diam}(G) \leq 2 \text{rad}(G)$.*

Proof. The first inequality follows from the definitions. Let u and v be vertices with $d(u, v) = \text{diam}(G)$, and let w be a vertex with $e(w) = \text{rad}(G)$. By the triangle inequality,

$$\text{diam}(G) = d(u, v) \leq d(u, w) + d(w, v) \leq 2e(w) = 2 \text{rad}(G). \quad \square$$

Given a result like $\text{diam}(G) \leq 2 \text{rad}(G)$, it is natural to ask whether this bound is as good as possible. If it were the case that $\text{diam}(G) \leq 2 \text{rad}(G) - 1$, we say that the bound could be strengthened. We say that a bound in a theorem is **attained** if there is a graph for which it is an equality (preventing it from being strengthened). A bound is **sharp** if there are infinitely many graphs for which it is an equality. Note that for paths of odd order, $\text{rad}(P_{2k+1}) = k$ and $\text{diam}(P_{2k+1}) = 2k$. This infinite class shows that the bound is sharp. Graphs that make a bound an equality are called **extremal graphs** for the bound. For some bounds, it is possible to characterize the extremal graphs, thereby improving the bound. Graph theory bounds, sharpness, and extremal graphs are explored in detail in Section 4 of the Appendix.

If a graph has a large diameter, it has relatively few edges, so it follows that its complement has many edges and a small diameter.

Proposition 1.42. *If $\text{diam}(G) \geq 4$, then $\text{diam}(\overline{G}) \leq 2$.*

Proof. Let $\text{diam}(G) \geq 4$, and let x and y be vertices with $d(x, y) = 4$. If u and v are nonadjacent in G , then they are adjacent in $\overline{G}$. If u and v are adjacent in G , then x and y cannot both be in the neighborhoods of u and v , since that would imply an $x - y$ path of length at most 3. Thus u and v are both nonadjacent to either x or y . Then $d(u, v) = 2$ in $\overline{G}$. $\square$

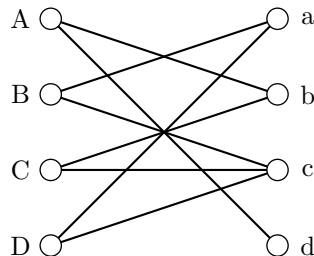
1.7. Bipartite Graphs

Some graphs have vertex sets that are divided into two distinct types of objects.

Example. A small college offers four math courses in the summer and has four math teachers who can each teach one class. We model this using a graph with four vertices representing courses and another four representing teachers. An edge joins a course to a qualified teacher. Note that it would make no sense for an edge to join two teachers or two classes.

The four teachers and the classes they are qualified to teach are given in the table. The situation is modeled with a graph at right.

Teacher	Classes
Alice	biomath, diff eq
Betty	algebra, calculus
Cindy	biomath, calculus
Dot	algebra, calculus



The natural question is whether it is possible to assign each teacher a course she is qualified to teach. Some thought shows that it is. Since only Alice can teach diff eq, Cindy must teach biomath, and Betty and Dot can choose between algebra and calculus. Thus there is a way to match each teacher with a class.

Definition 1.43. A graph is **bipartite** if the vertex set can be partitioned into two independent sets (called **partite sets**). A **matching** is a set of edges, none of which are adjacent. A **perfect matching** includes an endpoint of every vertex.

Bipartite graphs are also called 2-colorable since the two partite sets can be colored so that no adjacent vertices have the same color. A complete bipartite graph has all possible edges between partite sets. Paths, even cycles, grids, and hypercubes are other classes of bipartite graphs. Matchings in bipartite graphs are explored in Section 7.1.

The concept of distance is essential to the characterization of bipartite graphs.

Theorem 1.44. (Konig [1936]) *A graph is bipartite if and only if it contains no odd cycles.*

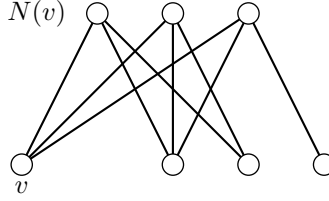
Proof. ($\Rightarrow$) Let G be a bipartite graph. Every walk alternates between the two partite sets, so it can only return to the first set after an even number of steps. Thus any cycle has even length.

($\Leftarrow$) Let G be a graph with no odd cycles. We assume G is connected, as this argument can be repeated for each component. Let u be a vertex, let U be the set of all vertices with even distance from u , and let W be the set of all vertices with odd distance from u . This is a partition of the vertex set. Suppose to the contrary that there is an edge between vertices x and y in the same partite set. Then the $u - x$ and $u - y$ geodesics have the same parity. These paths may have vertices in common, but there must be a last vertex v they have in common. Then the $v - x$ and $v - y$ geodesics and the edge xy can be combined to form an odd cycle. This is a contradiction, so U and W are independent sets. $\square$

Note that this is a constructive proof, as it provides a simple way to find the partite sets of a bipartite graph or show that it is not bipartite. This theorem is a **forbidden subgraph characterization**, where the class of graphs is characterized by a class of subgraphs that they do not contain. Such characterizations tell us about the structure of the graph class under consideration and may lead to practical algorithms for testing whether a graph is in the class. These and other graph characterizations are explored in Section 5 of the Appendix.

Bipartite graphs are useful in solving many problems. They are essential in the following result on triangle-free graphs.

Theorem 1.45 (Mantel [1907]). *The maximum number of edges in a triangle-free graph of order n is $\left\lfloor \frac{n^2}{4} \right\rfloor$, and the unique extremal graph is $K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}$.*



Proof. Let G be a graph with order n containing a vertex v with maximum degree d . None of v 's neighbors are adjacent, so each edge is incident with a vertex not in $N(v)$. Then $m(G) \leq \sum_{u \notin N(v)} d(u) \leq d(n-d) = m(K_{d,n-d})$ since we sum over $n-d$ vertices with degree at most d . To maximize the size of $K_{d,n-d}$, we note that moving a vertex from the partite set of size d to the other one adds $d-1$ edges and subtracts $n-d$ edges. The net gain of $2d-n-1$ is positive when $d > \frac{n+1}{2}$ and negative when $d < \frac{n+1}{2}$. Then the size is maximized when d is $\lfloor \frac{n}{2} \rfloor$ or $\lceil \frac{n}{2} \rceil$, so the maximum is $\lfloor \frac{n^2}{4} \rfloor$.

Equality requires that G is bipartite with $n-d$ vertices of degree d , and these numbers are as close as possible. Thus $G = K_{\lfloor \frac{n}{2} \rfloor, \lceil \frac{n}{2} \rceil}$. $\square$

It is interesting that the number of edges required to guarantee a triangle is the same number required to guarantee any odd cycle.

1.8. Generalizations of Graphs

There are many models related to, but different from, graphs as we have defined them. Recall the definition of a graph.

Definition 1.46. A **graph** G is a mathematical object consisting of a finite nonempty set of objects called **vertices** $V(G)$ (the **vertex set**), and a set of **edges** $E(G)$ (the **edge set**). An edge is two-element subset of the vertex set.

Consider how changing a few words in the definition of a graph results in different models. What if we remove “finite” or “nonempty”?

Definition 1.47. The **null graph** has an empty vertex and edge set.

There isn't much to do with the null graph. It is an exception to some theorems whose hypotheses it would vacuously satisfy. It is sometimes convenient to consider in counting formulas, but the complications outweigh the benefits, so we do not consider it a graph. These issues are considered further in Frank Harary and Ronald Read's [1974] tongue-in-cheek paper, *Is the Null Graph a Pointless Concept?*

Definition 1.48. An **infinite graph** has an infinite vertex set. A **finite graph** is not infinite.

There are many substantial results on infinite graphs, but it is better to have a firm understanding of finite graphs before considering them. Some results on finite graphs extend naturally to infinite graphs, but others do not. All graphs in this book will be finite unless explicitly stated.

What if the edges are not 2-element subsets?